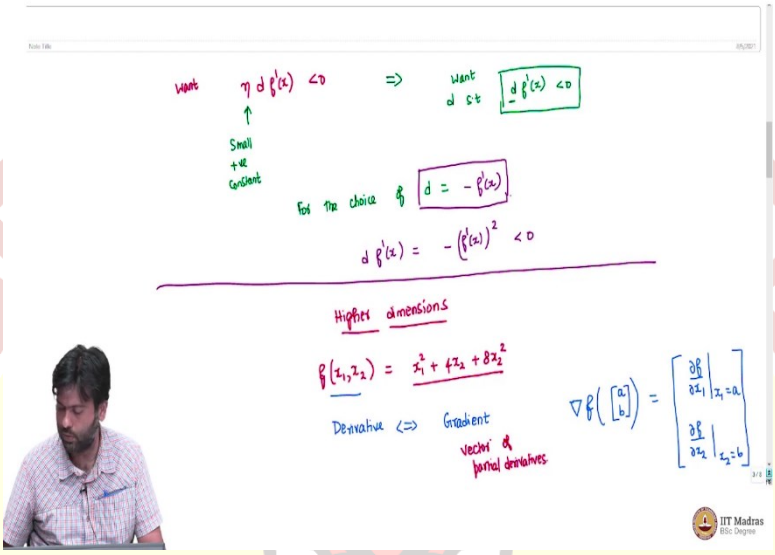


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Gradient Descent for Multivariate Functions

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Now, what we are going to look at is we are going to kind of generalize this idea of moving in the negative derivative's direction to functions of multiple variables. So, we want to now look at higher dimensions. What do I mean by higher dimensions? Well, now the function is not a function of a single variable, but the function is a function of multiple variables.

So, for example, the function could be of the following form; f of x 1 , x 2 could be something like maybe x 1 square + 4 x 2 + 8 x 2 squared. So, that could be, it could be some arbitrary function which is which depends on two variables. So now, the question is how do I you know, look at the corresponding ideas that we looked at in this particular case.

So, in other words, what is the equivalent of a derivative in this higher dimension, in the higher dimensions, the derivative will now we will think of the derivative in higher dimensions as what we will call as the gradient. Now what is a gradient, well, if you have a function like this, what you can do is, what does derivative capture, the derivative simply captures for a 1 dimensional function, the rate of change with respect to a particular point, so the moving in a particular direction.

So, the gradient is going to capture this for moving in every possible direction in x1's direction x2's direction, if it is a function of two variables, it is going to capture it in two different

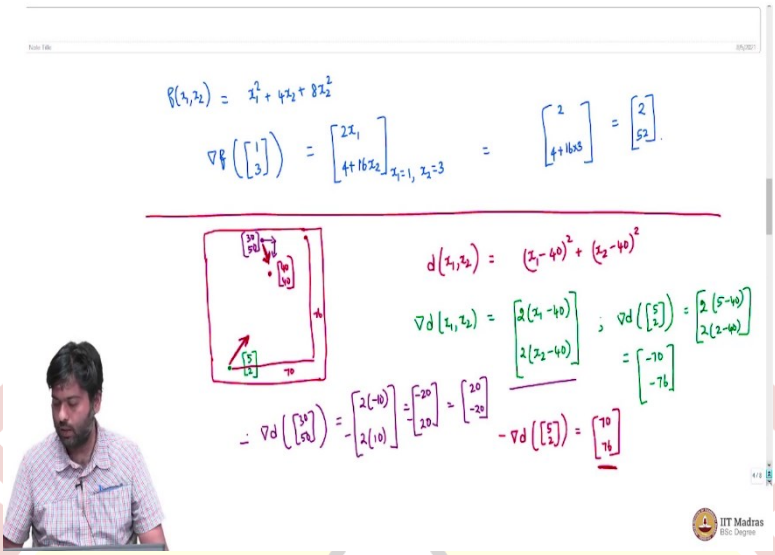
directions, in particular, the gradient, what kind of an object is the gradient? The gradient is a vector of partial derivatives, instead of the derivative, we will now look at the partial derivatives. What do I mean by that? I mean the following.

So, you have a function in this case, the function is a function of two variables x 1 and x 2, now what we are going to do is we are going to fix one of these values and see how the function behaves or how the derivative of the function behaves with respect to the other value. So, if you treat x 1 as constant, then you get the partial derivative of the function with respect to x 2. And similarly, if you fix x 2 as constant, then you get the partial derivative with respect to x 1. Now, the gradient then is just collecting these into a vector.

So, it is, it is a vector, maybe I will write this first. So, it is a vector of partial derivatives, let us say you are trying to compute the gradient and can maybe give the definition the, the notation first. Let us say you want to compute the gradient and it is typically use, uses this symbol nabla. So, δ f that is the instead of f ‘, we are going to call it as δ f, which is the gradient of the function, let us say at some particular point. Now, at every point is a vector in 2 dimension for, for a 2 dimensional function.

So let us say the point is some a , b, this is where we want to understand what the gradient looks like. So now this is going to be a 2 dimensional vector of partial derivatives, which is dou f by dou x 1 now evaluated at x 1 equals a and δdou f by dou x 2 evaluated at x 2 equals b. So this would be the definition of the gradient of the function f at the point a , b. Now, for example, if you wanted to compute the gradient for this particular point that we already have, let me go back to the function x 1 squared + 4 x 2 + 8 x 2 squared.

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Let me make sure you the same function, x 1 squared + 4 x 2 + 8 x 2 squared, so this is the function. Now, if I asked the question, let us say what is the gradient of this function at a point, let us say 1 , 3, so how would I calculate that, I would simply take the partial derivatives with respect to x 1, which will give me 2 x 1, and with respect to x 2 would give me 4 + 16 x 2. So, this is the general gradient at any x 1 x 2 and now we want to evaluate it at x 1 equals 1 and x 2 equals 3, which for which we will get 2 and 4 + 16 × 3, which is 52. So, this is going to be 2 , 52.

Remember again, so this is a vector, so it is going to give you, and that is why we call it a direction, so vector has both direction and magnitude. And so, whenever it say, say we move in the direction of the negative gradient, so it means that we are moving in a particular direction, which is given by this way, that is what it means.

So just let me do one other simple example, so that we are kind of clear here, we remember, so we were looking at this cow and grass example, where we wanted to find the distance from the grass, which was at a point 40 , 40. So, this was the point where the grass was, let me make this slightly bigger.

So, we wanted to find the distance to the cow from the grass, which was at 40, 40. And we said that that distance function, which we call this d, was again a function of 2 variables, d of x 1, x 2, and it was, well, x 1 - 40 squared + x 2 - 40 squared. This is the length of this point, length squared of the point from x 1 x 2 to 40 , 40, which is where the grass was.

So now let us see what intuitively the gradient is trying to do here, so let us say we are at a current point, which is, in this case, let me say, simply 5 , 2. Let us say we are at this point, so if I am at this point, well, I know that the minimum distance to the grass happens at 40 , 40. Now let us say I am at 5 , 2, and then I want to compute the gradient of this function at 5 , 2. So, what is the gradient of d, well, that is going to be at x 1 x 2 is going to be, well, if I take the partial derivative with respect to x 1, I get 2 x 1 - 40.

And if I take the partial derivative with x 2, I get 2 x 2 - 40. So, this will be my gradient. Now, I am going to try to evaluate it at 5 , 2, so 5 2 this point. Now, what is this gradient, well, of course, I can substitute the value, so it is going to be 2 into 5 - 40 2 into 2 - 40. So, it is going to be 2 × - 35 which is - 70. And 2 × - 38, I guess, which is - 76.

Now, remember, our gradient descent would tell us that we need to move in the negative of the gradient direction, which means that if I am at 5, 2, my updated step is going to be in the negative of the gradient direction - d of 5 , 2, which is in this case, 70, 76. Now, the numbers are not important.

But the point is that the - δ d of 5 , 2 is 70, 76 both the coordinates are positive, which means that if I am at this point, and I am going to add, I mean, what does it mean to say I am moving along this direction, I want to add some multiple of this vector, which as, which is eta which is a small number. So, which means I am going to take a small step in the direction of 70 , 76.

But what does, what does 70 , 76 mean, so it is going to be 70, , 76. So, 70 in this direction, 76 in this direction. So, which means that I am going to move in this direction. Of course, I would not move all the way, by adding 70, 76, I am going to take very, very small step eta step. So, which means that I am the point is, I am going to move in this direction.

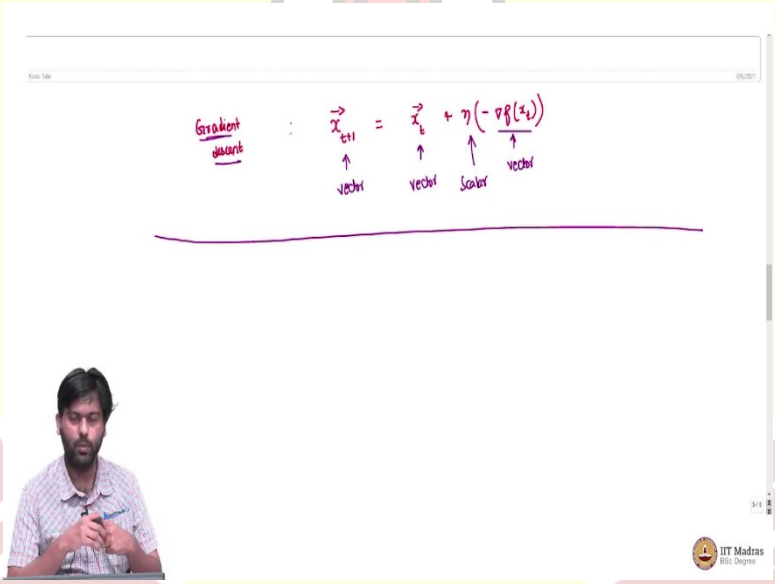
Well, that pictorially tells us that if I move in this direction, I am going to get closer to 40 , 40. So, because in this direction, my function actually decreases. And so, I am getting closer to 40 , 40. And this is not really a coincidence, so you can, for instance, ask at a different point. Well, let us say in this case, I look at a point which is somewhere here 30 , let us say 50, which is a point here, maybe not there. Maybe on this side.

If we look at the point 30 , 50 and now I compute what is the derivative at 30 , 50. Well, again, I substitute it in the equation of 2 x 1 - 40 2 x 2 - 40. What would I get, I would get 2 × - 10 2 × 10 which is - 20 and 20 in this particular case, so happens, which means that the negative of the gradient is going to be the negative of this, which is 20 and - 20.

So, which means that in the x direction, I am moving in a positive direction in the y direction, I am moving it in the negative direction, which means I am going to move the x in this direction, y in this direction, which means I will move in the, in the towards the third quadrant in this case, and that is, that will be my direction of movement let me highlight it in red. So, my next point is going to be along this direction. And yes, you can see again that this is also trying to get me closer to 40 , 40.

So, the idea is that the gradient will always the negative of the gradient will always point us towards direction of decrease. And in this case, it turns out that it is always pointing to the place where the minimum is. So, which means that we can hope that as we travel along this, of course, as I go in this direction, my gradient direction might potentially change depending on the function, but then it will always move such that eventually I will reach the minimum. That is the general idea of you know, generalizing simple, the derivative base thing to a gradient.

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So, which means let me finally say that in the gradient descent algorithm for higher dimensions, let me make that formal is going to have the following update role x t + 1. Now remember, x is a vector, it is a vector depending on the number of dimensions or the number of variables in your problem equals x t, you start at some point and then you take an eta step, eta is still a scalar, it is just step size × - δ f of x t.

So, this is a vector, this is a vector, this is a vector, this is a scalar. So, now, this is a generalization of our simple algorithm, which was just taking the negative f ‘ of x to - δ f of x t. So, because once we have generalized derivatives to a vector of partial derivatives, we can simply apply the same algorithm.

And because this is gradient, the general term for this gradient algorithm is called gradient descent. So, this is the, this algorithm now works for any unconstrained optimization problem as long as you can compute the gradient of the function, you can keep moving along the gradient direction and of course, this will take you to a local minimum. So, we discussed local minimum earlier. So, this function, this algorithm is guaranteed to converge to a local minimum.